香港中文大學（深圳）
The Chinese University of Hong Kong, Shenzhen

Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	CSC1002
Course title (English)	Computational Laboratory
Course title (Chinese)	计算机实验
Units	1
Language of Instruction	English
Description (English)	This course prepares students in acquiring hands-on software programming skills using Python and other related development tools and libraries as needed. Students will be given a number of programming assignments in increasing complexity that covers many essential aspects of software development using fundamental data structures, functions, modules and classes.
Description (Chinese)	此课程帮助学生通过实际使用 Python 编程语言和其他相关的开发工具和库来更好地掌握软件编程技能。课程将涵盖不同难度和类别的编程任务，需要使用数据结构、函数、模块和类等编程概念。

2. Prerequisites / Co-requisites

A. Prerequisites

CSC1001

B. Co-requisites

3. Learning Outcomes

Knowledge:

• Students will be introduced to computer programming with focus on fundamental techniques utilizing simple implementation methods & approaches.

• Students will be introduced to the concept of fundamental Software Development Life Cycle: analysis, design, coding, testing and documentation.

• Students will gain an understanding of the fundamental programming elements and the basic principles of modular design in software application development.

• Students will learn the pitfalls and tips as well as best practices in software programming.

Skill (Generic):

• Students will be able to install and configure a programming development platform based on python.

• Students will be familiar with general language syntax and programing using an IDE.

• Students will be able to design and develop computer applications using standard programming elements such as list, array, tuple, dictionary, class, function, string and so on.

• Student will gain hands-on skills in software development including designing, coding, troubleshooting and debugging.

• Student will understand and apply best practices in programming in naming convention, coding styles, documentation, and design framework such as separation of concerns and single responsibility principle for code reuse and simplicity.

Value:

• Students will be equipped with general programming skills that they can directly apply to many other areas handily as needed for the benefit of users.

• Students will have awareness of the values & practical uses of programming and might decide to continue to advance their knowledges in computer science in the pursuit of computer scientist or computer profession in software engineering.

• Students will be more literate and vigilant in programming so that they will integrate and/or combine such knowledge in the field of other studies, be it science or non-science, in a creative and innovative manners.

4. Course syllabus

This course will introduce the many aspects of programming concepts using Python as the programming language through a series of lab exercises, programming assignments and group projects, and it will explain the basics of structured programming using functions, classes and modules. This course will form a sound basis for further study of computer science.

The main topics covered in this course are:

•Hands-on programming using Python language;
•Data types and operators in Python language;
•Input/output; Flow control and loop; Function; List;
•Software Design
•Problem Decomposition

The mark for an assessment item submitted after the designated time on the due date, without an approved extension of time, will be reduced by 10% of the possible maximum mark for that assessment item for each day or part day that the assessment item is late. Note: this applies equally to week and weekend days.

5. Assessment Scheme

Component/ method	**% weight**
Programming Assignment I, II, III (30%, 30%, 40% RESPECTIVELY)	100
OR Programming Assignment I, II, Mid-Term Quiz (30%, 40%, 30% RESPECTIVELY)	100

6. Grade descriptor

General

Grade	**Description**
A	Demonstrates the ability to achieve all the learning outcomes in this course and be able to follow all "Best-Practice" in accordance to "Assessment Scheme". Students shows outstanding performance that surpass the normal expectation at this level with exceptional skills in creativity and innovation, both in class and take-home assignments
A-	Mostly follow "Best-Practice" in accordance to "Assessment Scheme" with minor, non-critical errors found. Students demonstrates strong ability in understanding and applying all "Best-Practice" requirements in a clear and best manner
B	Generally follow "Good" requirements in accordance to "Assessment Scheme" with several non-critical errors. Students show strong ability in both "Best-Practice" & "Good" areas with setbacks and has the ability to apply the knowledge in a satisfactory and unambiguous manner
C	Generally follow "Good/Average" requirements with several errors including critical ones. Students show general understanding of the principles learnt in the course in a manner that is not incorrect but is somewhat fragmented.
D	Generally follow "Average/Poor" requirements with several errors including critical

	ones. Students show basic understanding of the principles learnt in the course in a manner that is broadly correct in its essential in some simple and familiar situations.
F	Unsatisfactory performance on a number of learning outcomes; OR failure to meet majority of assessment requirements.

7. Feedback for evaluation

Formal Course and Teaching Evaluation
Informal feedback to instructor and/or teaching assistant

8. Reading

9. Course components

Activity	Hours/week
Labs	1
Tutorial	2

10. Indicative teaching plan

Week	Content/ topic/ activity
001	Python quick overview; learning outcomes and OBA
002	Live Coding – Largest 3 (basic construct, refactoring, functions, generalization), !!! Eval
003	Coding Styles – Naming Convention, Layout, Variable Scope, Functions, Comments Assignment 1 (User Interface/Console)
004	Live Coding – Caesar Cipher (functions, loop, layout, controls)
005	Live Coding – Simple board game (flipping number) – (Scope, Spec, Design, Implementation) Program structure – Console Base
006	Introduction to Turtle Graphics (Building blocks – canvas, turtles, pen, draw, motion)
007	Live Coding – Simple board game (flipping number) – GUI Implementation Part 1
008	Live Coding – Simple board game (flipping number) – GUI Implementation Part 2 Assignment 2
009	More Turtle Graphics (Screen, Timer, Click, Events, Mouse Tracking) Part 1
010	More Turtle Graphics (Screen, Timer, Click, Events, Mouse Tracking) Part 2

011	Assignment 3
012	Assignment 2 - Q&A, Other programming tools: Debugging, Source Control, Virtual Environment
013	Review

11. Any other information

12. Version date

Version number	2
As of (date)	2023/11/22